

CSE225:DEVELOPING ANDROID APPS

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: develop Android applications using Jetpack Compose UI components and Material design elements.

CO2 :: apply background tasks and inter-component communication using Intents, WorkManager, and Notifications.

CO3 :: create composable screens and handle in-app navigation using Navigation Compose.

CO4 :: create reusable and customized user interfaces using custom composables, modifiers, and Material theming

CO5 :: apply secure data storage and access techniques using DataStore, Scoped Storage, and runtime permissions.

CO6 :: demonstrate effective single-activity navigation using tabs, pagers, and navigation drawer components

Unit I

Views & Components in Jetpack Compose : Lists and grids, Scroll and Nested Scroll, Adding Spinner to App, Splash using LaunchedEffect, Progressbar, Custom Rating Bar, Splash Screen, Menu

Unit II

Managing Application Communication and Scheduling : Intents, Pending Intents, Comparing Intents and Pending Intents, Alarm Manager, Job Scheduler, Notification, Implementing notification channels

Unit III

Notifications and User Interaction : Notification, Implementing notification channels, Date Picker Dialog, Time Picker Dialog

Unit IV

Custom UI Components in Jetpack Compose : Material & Foundation composables, fully declarative UI approach, Creating custom Text, TextField, Button using parameters, Modifier, Shape, Color

Unit V

Modern Data Storage & Permission Management : DataStore (Preferences DataStore), Scoped Storage, Runtime Permissions with Compose UI

Unit VI

Advanced Navigation & Paging UI Components in Jetpack Compose : TabRow + HorizontalPager, ModalNavigationDrawer / NavigationDrawer, HorizontalPager / VerticalPager

List of Practicals / Experiments:

List of Practical

- Create an application based on different Views & Components.
- Create an application based on Intents & Different Schedulers.
- Create an application to implement Custom Views.
- Create an application based on different Storage Options.
- Create an application to implement different types of Navigation.
- Build an app to implement UI Components and Basic Views
- Build an app to implement intents & background scheduling
- Build an app to use fragments and data transfer
- Build an app for custom views and advanced UI elements
- Build an app to explore data storage techniques
- Build an app to implement effective navigation

Text Books:

1. PROFESSIONAL ANDROID 4 APPLICATION DEVELOPMENT by RETO MEIER, WILEY

References:

1. JETPACK COMPOSE 1.7 ESSENTIALS by NEIL SMYTH, Payload Publisher